

```
User {  
  user_id PK  
  email UK  
  username UK  
  password_hash  
  phone  
  avatar_url  
  cover_image_url  
  bio  
  headline  
  country_code  
  timezone  
  preferred_language  
  languages_spoken  
  social_links JSON  
  tokens_balance  
  level  
  experience_points  
  reputation_score  
  is_email_verified  
  is_phone_verified  
  is_teacher_verified  
  is_premium_member  
  status ENUM(active, suspended, banned, deleted)  
  role ENUM(user, teacher, admin, super_admin)  
  created_at  
  updated_at
```

```
last_login_at
last_active_at
}
TeacherProfile {
  teacher_profile_id PK
  user_id FK > User.user_id UK
  verification_status ENUM(pending, verified, rejected)
  verification_documents JSON
  hourly_rate
  available_for_hire
  years_experience
  company
  title
  education JSON
  certifications JSON
  portfolio_url
  linkedin_url
  github_url
  website_url
  intro_video_url
  total_hours_taught
  student_count
  completion_rate
  response_rate
  response_time_minutes
  created_at
  updated_at
```

}

LearnerProfile {

learner_profile_id PK

user_id FK > User.user_id UK

learning_goals JSON

preferred_learning_style ENUM(visual, auditory, kinesthetic, reading)

skill_level ENUM(beginner, intermediate, advanced, expert)

budget_range_min

budget_range_max

time_availability JSON

preferred_teacher_types JSON

created_at

updated_at

}

UserPreference {

preference_id PK

user_id FK > User.user_id UK

notification_settings JSON

privacy_settings JSON

communication_preferences JSON

language_preferences JSON

accessibility_settings JSON

theme ENUM(light, dark, system)

time_format ENUM(12h, 24h)

date_format

created_at

```
    updated_at
}
```

```
# =====
# SKILL & CATEGORY SYSTEM
# =====
```

```
SkillCategory {
  category_id PK
  name UK
  slug UK
  description
  icon
  parent_category_id FK > SkillCategory.category_id
  display_order
  is_active
  created_at
  updated_at
}
```

```
Skill {
  skill_id PK
  name UK
  slug UK
  description
  short_description
  category_id FK > SkillCategory.category_id
```

```
subcategories JSON
difficulty ENUM(beginner, intermediate, advanced, expert)
avg_completion_hours
prerequisites JSON
learning_outcomes JSON
tags JSON
featured_image_url
thumbnail_url
popularity_score
demand_score
avg_hourly_rate
is_featured
is_verified
is_active
created_by FK > User.user_id
created_at
updated_at
}
```

```
UserSkill {
  user_skill_id PK
  user_id FK > User.user_id
  skill_id FK > Skill.skill_id
  proficiency ENUM(beginner, intermediate, advanced, expert)
  years_experience
  teaching_level ENUM(can_teach, expert_teacher)
  learning_level ENUM(want_to_learn, currently_learning)
```

```
verification_status ENUM(self_claimed, verified, pending_verification)
verification_method ENUM(portfolio, certification, test, experience)
is_primary_skill
hourly_rate
availability_status ENUM(available, limited, not_available)
student_count
rating
review_count
created_at
updated_at
}
```

```
SkillBadge {
  badge_id PK
  skill_id FK > Skill.skill_id
  name
  description
  icon
  requirements JSON
  xp_reward
  token_reward
  is_active
  created_at
}
```

```
UserSkillBadge {
  user_badge_id PK
```

```
user_id FK > User.user_id
badge_id FK > SkillBadge.badge_id
awarded_at
expires_at
is_active
}
```

```
# =====
# SESSION & BOOKING SYSTEM
# =====
```

```
Session {
  session_id PK
  session_code UK
  teacher_id FK > User.user_id
  learner_id FK > User.user_id
  skill_id FK > Skill.skill_id
  user_skill_id FK > UserSkill.user_skill_id
  title
  description
  objectives JSON
  prerequisites JSON
  materials_required JSON
  session_type ENUM(one_on_one, group, workshop, course)
  session_format ENUM(live, recorded, hybrid)
  difficulty_level ENUM(beginner, intermediate, advanced)
  max_participants
```

scheduled_start_time
scheduled_end_time
duration_minutes
actual_start_time
actual_end_time
status ENUM(draft, published, scheduled, in_progress, completed, cancelled,
no_show, rescheduled)
cancellation_reason
cancellation_initiated_by ENUM(teacher, learner, admin, system)
is_recurring
recurrence_pattern JSON
parent_session_id FK > Session.session_id
created_at
updated_at
}

SessionBooking {
booking_id PK
session_id FK > Session.session_id
learner_id FK > User.user_id
booking_code UK
status ENUM(pending, confirmed, cancelled, waitlisted, refunded)
booking_time
cancellation_time
cancellation_reason
notes
special_requirements JSON


```
    created_at
    updated_at
}
```

```
SessionParticipant {
    participant_id PK
    session_id FK > Session.session_id
    user_id FK > User.user_id
    role ENUM(teacher, learner, assistant, observer)
    join_time
    leave_time
    attendance_status ENUM(present, absent, late, left_early)
    participation_score
    created_at
}
```

```
# =====
# VIDEO & MEDIA SYSTEM
# =====
```

```
VideoSession {
    video_session_id PK
    session_id FK > Session.session_id UK
    agora_channel_name
    agora_token
    agora_uid
    recording_resource_id
}
```

```
    recording_sid
    recording_status ENUM(not_started, recording, stopped, uploaded, processing, ready,
failed)
    recording_start_time
    recording_end_time
    video_url
    playback_url
    thumbnail_url
    duration_seconds
    file_size_bytes
    storage_provider ENUM(agora, s3, cloundinary, vimeo)
    is_public
    view_count
    created_at
    updated_at
}
```

```
WhiteboardSession {
    whiteboard_id PK
    session_id FK > Session.session_id
    whiteboard_data JSON
    snapshot_url
    created_at
    updated_at
}
```

```
ScreenShare {
```

```
screen_share_id PK
session_id FK > Session.session_id
user_id FK > User.user_id
start_time
end_time
recording_url
created_at
}
```

```
# =====
# AI & ANALYTICS SYSTEM
# =====
```

```
SessionTranscript {
  transcript_id PK
  session_id FK > Session.session_id UK
  raw_text
  processed_text
  speaker_labels JSON
  timestamps JSON
  word_count
  language
  confidence_score
  created_at
  updated_at
}
```

```
AI_SessionSummary {  
  summary_id PK  
  session_id FK > Session.session_id UK  
  transcript_id FK > SessionTranscript.transcript_id  
  ai_model  
  summary_text  
  key_points JSON  
  action_items JSON  
  recommendations JSON  
  engagement_score  
  teaching_effectiveness_score  
  learning_outcomes_assessment JSON  
  sentiment_analysis JSON  
  topics_discussed JSON  
  generated_at  
  created_at  
  updated_at  
}
```

```
AI_Recommendation {  
  recommendation_id PK  
  user_id FK > User.user_id  
  skill_id FK > Skill.skill_id  
  recommendation_type ENUM(skill_to_learn, teacher_to_follow, session_to_book,  
  content_to_watch)  
  reason  
  confidence_score
```

```
priority
is_viewed
is_accepted
created_at
}
```

```
# =====
# MESSAGING & COMMUNICATION
# =====
```

```
Conversation {
  conversation_id PK
  conversation_type ENUM(direct, group, support, session_group)
  title
  avatar_url
  is_active
  is_archived
  last_message_id FK > Message.message_id
  created_at
  updated_at
}
```

```
ConversationParticipant {
  participant_id PK
  conversation_id FK > Conversation.conversation_id
  user_id FK > User.user_id
  role ENUM(admin, member, muted, blocked)
```

```
joined_at
left_at
last_read_at
unread_count
notification_preferences JSON
}
```

```
Message {
  message_id PK
  conversation_id FK > Conversation.conversation_id
  sender_id FK > User.user_id
  message_type ENUM(text, image, video, file, system, session_alert)
  content
  metadata JSON
  attachments JSON
  is_edited
  edited_at
  is_deleted
  deleted_at
  deleted_by
  read_by JSON
  delivered_to JSON
  created_at
}
```

```
MessageReaction {
  reaction_id PK
```

```
message_id FK > Message.message_id
user_id FK > User.user_id
reaction_type ENUM(like, love, laugh, wow, sad, angry, celebrate, question)
created_at
}
```

```
# =====
# PAYMENT & TOKEN ECONOMY
# =====
```

```
PaymentMethod {
  payment_method_id PK
  user_id FK > User.user_id
  provider ENUM(stripe, paypal, bank_transfer, crypto)
  method_type ENUM(card, bank_account, wallet, crypto_wallet)
  provider_customer_id
  provider_payment_method_id
  is_default
  metadata JSON
  is_active
  created_at
  updated_at
}
```

```
Payment {
  payment_id PK
  user_id FK > User.user_id
```

session_id FK > Session.session_id
payment_method_id FK > PaymentMethod.payment_method_id
payment_intent_id UK
amount
currency
status ENUM(pending, processing, requires_action, succeeded, failed, canceled, refunded)
description
receipt_url
refund_reason
refunded_amount
metadata JSON
created_at
updated_at
}

TokenPackage {
package_id PK
name
description
token_amount
price
currency
discount_percentage
is_featured
is_active
display_order


```
    validity_days
    created_at
    updated_at
}
```

```
TokenTransaction {
    transaction_id PK
    user_id FK > User.user_id
    session_id FK > Session.session_id
    payment_id FK > Payment.payment_id
    transaction_type ENUM(purchase, session_earning, session_spending, bonus,
referral, refund, adjustment)
    token_amount
    balance_before
    balance_after
    description
    metadata JSON
    created_at
}
```

```
TokenTransfer {
    transfer_id PK
    sender_id FK > User.user_id
    receiver_id FK > User.user_id
    session_id FK > Session.session_id
    token_amount
    status ENUM(pending, completed, failed, reversed)
```

```
transfer_reason
created_at
updated_at
}
```

```
# =====
# RATING & REVIEW SYSTEM
# =====
```

```
Review {
  review_id PK
  session_id FK > Session.session_id
  reviewer_id FK > User.user_id
  reviewee_id FK > User.user_id
  review_type ENUM(teacher_review, learner_review, session_review)
  rating
  title
  content
  strengths JSON
  areas_for_improvement JSON
  is_anonymous
  is_verified_booking
  helpful_count
  report_count
  status ENUM(pending, published, flagged, hidden, deleted)
  created_at
  updated_at
}
```

```
}
```

```
ReviewResponse {
```

```
  response_id PK
```

```
  review_id FK > Review.review_id
```

```
  responder_id FK > User.user_id
```

```
  content
```

```
  created_at
```

```
  updated_at
```

```
}
```

```
ReviewHelpful {
```

```
  helpful_id PK
```

```
  review_id FK > Review.review_id
```

```
  user_id FK > User.user_id
```

```
  created_at
```

```
}
```

```
# =====
```

```
# NOTIFICATION & ALERT SYSTEM
```

```
# =====
```

```
NotificationTemplate {
```

```
  template_id PK
```

```
  name UK
```

```
  title_template
```

```
  body_template
```

```
notification_type ENUM(email, push, in_app, sms)
category ENUM(session, payment, message, system, marketing)
variables JSON
is_active
created_at
updated_at
}
```

```
Notification {
  notification_id PK
  user_id FK > User.user_id
  template_id FK > NotificationTemplate.template_id
  title
  message
  data JSON
  notification_type ENUM(email, push, in_app, sms)
  priority ENUM(low, normal, high, critical)
  status ENUM(pending, sent, delivered, read, failed)
  read_at
  action_url
  expires_at
  created_at
  updated_at
}
```

```
NotificationPreference {
  preference_id PK
```

```
user_id FK > User.user_id UK
email_notifications JSON
push_notifications JSON
sms_notifications JSON
in_app_notifications JSON
quiet_hours_start
quiet_hours_end
created_at
updated_at
}
```

```
# =====
# GAMIFICATION & ACHIEVEMENTS
# =====
```

```
Achievement {
  achievement_id PK
  name UK
  description
  icon
  achievement_type ENUM(session_count, rating, streak, referral, skill_mastery)
  requirements JSON
  xp_reward
  token_reward
  badge_image_url
  rarity ENUM(common, rare, epic, legendary)
  is_active
```

```
    created_at
    updated_at
}
```

```
UserAchievement {
    user_achievement_id PK
    user_id FK > User.user_id
    achievement_id FK > Achievement.achievement_id
    unlocked_at
    progress
    is_unlocked
    created_at
    updated_at
}
```

```
Streak {
    streak_id PK
    user_id FK > User.user_id UK
    current_streak_days
    longest_streak_days
    last_activity_date
    streak_type ENUM(session, login, learning)
    created_at
    updated_at
}
```

```
Leaderboard {
```

```
leaderboard_id PK
leaderboard_type ENUM(weekly, monthly, all_time, by_skill)
skill_id FK > Skill.skill_id
period_start
period_end
rankings JSON
created_at
updated_at
}
```

```
# =====
# CONTENT & LEARNING MATERIALS
# =====
```

```
LearningPath {
  learning_path_id PK
  title
  description
  skill_id FK > Skill.skill_id
  difficulty ENUM(beginner, intermediate, advanced)
  estimated_hours
  skills_covered JSON
  prerequisites JSON
  is_published
  is_featured
  created_by FK > User.user_id
  created_at
}
```

```
    updated_at  
  }
```

```
Course {  
  course_id PK  
  learning_path_id FK > LearningPath.learning_path_id  
  teacher_id FK > User.user_id  
  title  
  description  
  curriculum JSON  
  sessions_count  
  total_duration_hours  
  price_type ENUM(free, paid, subscription)  
  price  
  enrollment_count  
  rating  
  is_published  
  created_at  
  updated_at  
}
```

```
Content {  
  content_id PK  
  course_id FK > Course.course_id  
  session_id FK > Session.session_id  
  content_type ENUM(video, article, quiz, assignment, resource)  
  title
```



```
description
content_url
duration_minutes
is_preview
display_order
created_at
updated_at
}
```

```
# =====
# SUPPORT & MODERATION
# =====
```

```
SupportTicket {
  ticket_id PK
  ticket_code UK
  user_id FK > User.user_id
  assigned_to FK > User.user_id
  category ENUM(technical, billing, account, session, safety)
  priority ENUM(low, medium, high, urgent)
  subject
  description
  status ENUM(open, in_progress, resolved, closed)
  resolution_notes
  first_response_time
  resolution_time
  created_at
}
```

```
    updated_at
}
```

```
TicketMessage {
    ticket_message_id PK
    ticket_id FK > SupportTicket.ticket_id
    sender_id FK > User.user_id
    message_type ENUM(user_message, agent_message, system)
    content
    attachments JSON
    is_internal_note
    created_at
}
```

```
Report {
    report_id PK
    reporter_id FK > User.user_id
    reported_user_id FK > User.user_id
    reported_session_id FK > Session.session_id
    reported_review_id FK > Review.review_id
    report_type ENUM(user, session, review, message, content)
    category ENUM(harassment, spam, inappropriate, fake, safety, other)
    description
    evidence JSON
    status ENUM(pending, investigating, resolved, dismissed)
    priority ENUM(low, medium, high, critical)
    assigned_to FK > User.user_id
}
```

```
resolution ENUM(warning, suspension, ban, content_removed, no_action)
resolution_notes
created_at
updated_at
}
```

```
# =====
# ADMIN & SYSTEM MANAGEMENT
# =====
```

```
AdminUser {
  admin_id PK
  user_id FK > User.user_id UK
  role ENUM(super_admin, admin, moderator, support, analyst)
  permissions JSON
  department ENUM(operations, support, content, finance, analytics)
  is_active
  last_login
  created_by FK > AdminUser.admin_id
  created_at
  updated_at
}
```

```
SystemSetting {
  setting_id PK
  key UK
  value JSON
}
```

```
data_type ENUM(string, number, boolean, array, object)
category ENUM(general, payment, session, email, security, features)
description
is_public
last_updated_by FK > AdminUser.admin_id
created_at
updated_at
}
```

```
AuditLog {
  log_id PK
  admin_id FK > AdminUser.admin_id
  user_id FK > User.user_id
  action_type
  entity_type
  entity_id
  old_values JSON
  new_values JSON
  ip_address
  user_agent
  created_at
}
```

```
# =====
# ANALYTICS & MONITORING
# =====
```

```
UserAnalytics {  
  analytics_id PK  
  user_id FK > User.user_id UK  
  date  
  sessions_booked  
  sessions_taught  
  tokens_spent  
  tokens_earned  
  messages_sent  
  logins  
  active_minutes  
  learning_progress JSON  
  created_at  
}
```

```
PlatformAnalytics {  
  platform_analytics_id PK  
  date  
  total_users  
  active_users  
  new_signups  
  total_sessions  
  completed_sessions  
  revenue  
  token_transactions  
  popular_skills JSON  
  user_retention_rate
```

```
    created_at
}
```

```
SessionAnalytics {
    session_analytics_id PK
    session_id FK > Session.session_id
    engagement_metrics JSON
    participation_metrics JSON
    technical_metrics JSON
    feedback_metrics JSON
    created_at
}
```

```
# =====
# RELATIONSHIP DEFINITIONS
# =====
```

```
# User Relationships
User has_one TeacherProfile
User has_one LearnerProfile
User has_one UserPreference
User has_many UserSkill
User has_many TokenTransaction
User has_many Payment
User has_many Notification
User has_many UserAchievement
User has_one Streak
```

User has_many SupportTicket

User has_many Report

User has_many Review

User has_many Message

User participates_in Conversation

User has_many Session as teacher

User has_many Session as learner

User follows_many User

User is_followed_by_many User

Skill Relationships

Skill belongs_to SkillCategory

Skill has_many UserSkill

Skill has_many LearningPath

Skill has_many Session

Skill has_many SkillBadge

Session Relationships

Session has_one VideoSession

Session has_one SessionTranscript

Session has_one AISessionSummary

Session has_one WhiteboardSession

Session has_many ScreenShare

Session has_many SessionParticipant

Session has_many Review

Session has_one SessionBooking

Session has_one Payment

Session belongs_to Skill

Session belongs_to User as teacher

Session belongs_to User as learner

Payment Relationships

Payment belongs_to User

Payment belongs_to Session

Payment belongs_to PaymentMethod

Payment has_one TokenTransaction

Conversation & Messaging

Conversation has_many Message

Conversation has_many ConversationParticipant

Message has_many MessageReaction

Content Relationships

LearningPath has_many Course

Course has_many Content

Course belongs_to User as teacher

Admin Relationships

AdminUser manages_many Report

AdminUser updates_many SystemSetting

AdminUser creates_many AuditLog

AdminUser assigned_many SupportTicket

Analytics Relationships

UserAnalytics belongs_to User

PlatformAnalytics is_aggregated_daily

SessionAnalytics belongs_to Session

Token System

TokenPackage is_purchasable_by User

TokenTransaction tracks User tokens

TokenTransfer between_users